

Assignment-19.1

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Batch:12

Task 1 – Python to Java Conversion

Task:

Translate a simple Python class into Java using AI assistance.

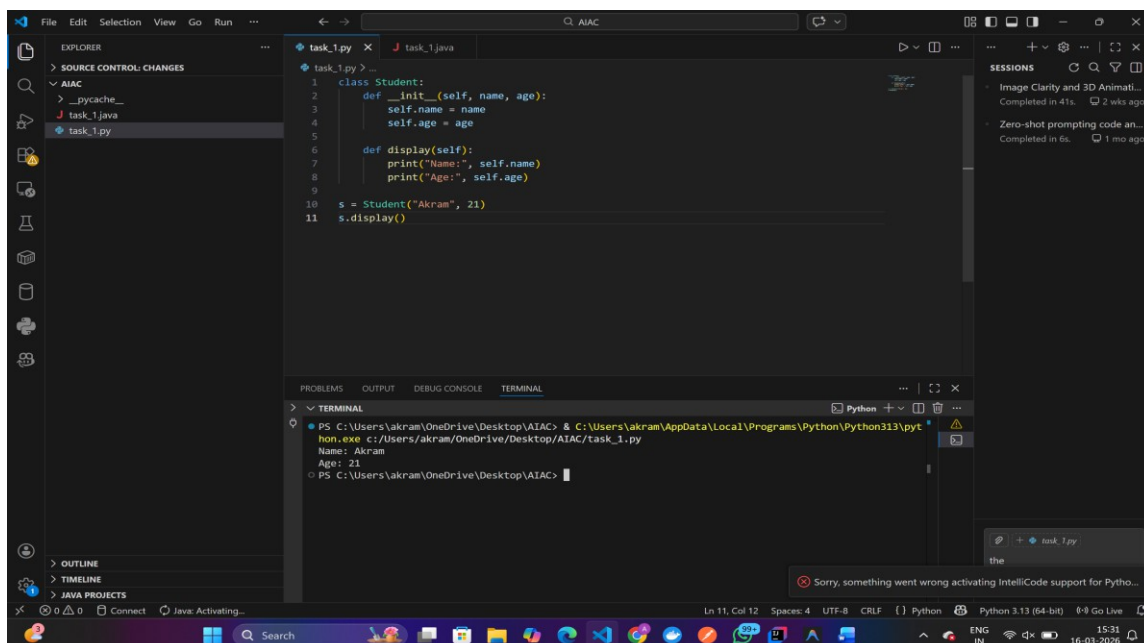
Instructions:

- Prompt AI to convert a Python class with attributes and methods into equivalent Java code.
- Verify that constructors, method signatures, and data types are properly handled.
- Run both versions (Python & Java) to compare expected outputs.

Expected Output:

- Equivalent working Java class translated from Python.

CODE:

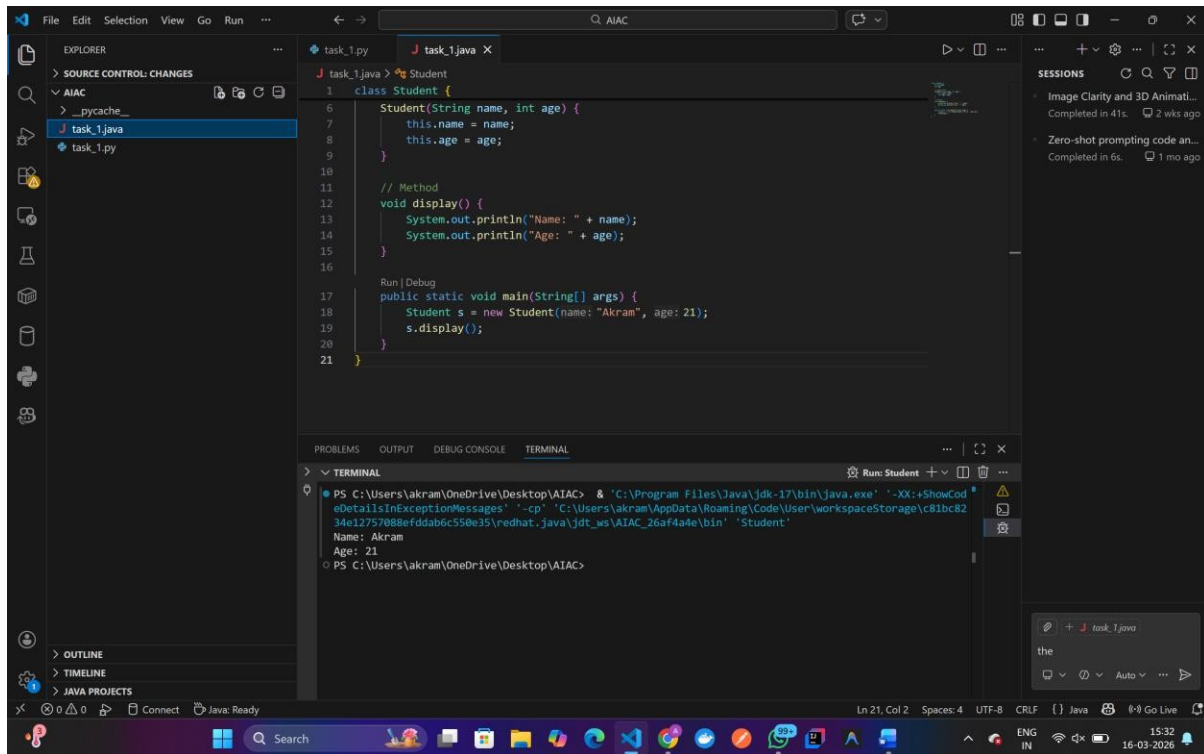


The screenshot shows a Visual Studio Code editor with a Python file named `task_1.py`. The code defines a `Student` class with an `__init__` method and a `display` method. The `display` method prints the name and age of the student. An instance of the class is created with the name 'Akram' and age 21, and the `display` method is called. The terminal output shows the execution of the script, resulting in the printed name and age.

```
1 class Student:
2     def __init__(self, name, age):
3         self.name = name
4         self.age = age
5
6     def display(self):
7         print("Name:", self.name)
8         print("Age:", self.age)
9
10 s = Student("Akram", 21)
11 s.display()
```

Terminal Output:

```
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:\Users\akram\OneDrive\Desktop\AIAC\task_1.py
Name: Akram
Age: 21
PS C:\Users\akram\OneDrive\Desktop\AIAC>
```



Task 2 – C++ to Python Function Conversion

Task:

Convert a C++ function for factorial calculation into Python.

Instructions:

- Ask AI to translate a recursive C++ factorial function into Python.
- Validate correctness by testing with sample inputs.
- Ensure Python version follows Pythonic conventions.

Expected Output:

- A correct Python implementation equivalent to the original C++ factorial function.

CODE:

The screenshot shows the Visual Studio Code editor with a file explorer on the left containing files like `task1.py`, `task2.py`, and `launch.json`. The main editor displays `task2.py` with the following Python code:

```

1 def factorial(n):
2     if n == 0:
3         return 1
4     else:
5         return n * factorial(n-1)
6
7 print(factorial(5))

```

The bottom panel shows the TERMINAL with the following output:

```

PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_1.py
Name: Akram
Age: 21
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task2.py
120
PS C:\Users\akram\OneDrive\Desktop\AIAC>

```

The screenshot shows the Visual Studio Code editor with a file explorer on the left containing files like `task1.py`, `task2.py`, and `task2.c++`. The main editor displays `task2.c++` with the following C++ code:

```

1 #include <iostream>
2 using namespace std;
3
4 int factorial(int n) {
5     if(n == 0)
6         return 1;
7     else
8         return n * factorial(n-1);
9 }
10
11 int main() {
12     cout << factorial(5);
13     return 0;
14 }

```

The bottom panel shows the TERMINAL with the following output:

```

PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_1.py
Name: Akram
Age: 21
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task2.py
120
PS C:\Users\akram\OneDrive\Desktop\AIAC>

```

Task 3 – SQL to Pandas Query

Task:

Translate a SQL query into a Pandas DataFrame operation in Python.

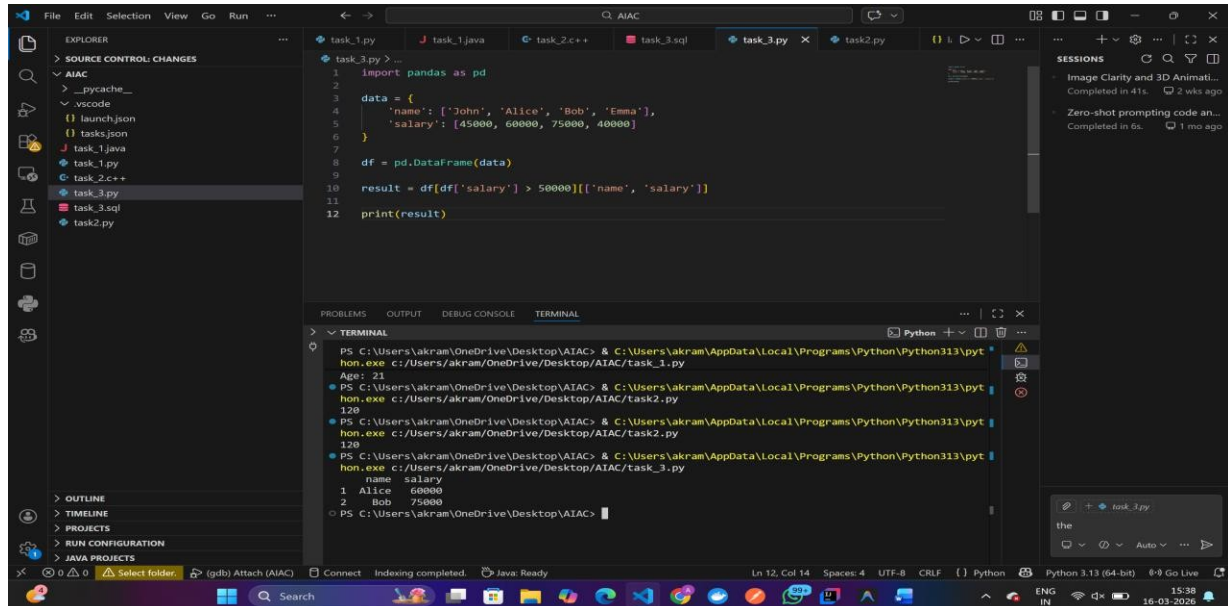
Instructions:

- Provide AI with a SQL query (e.g., SELECT name, salary FROM employees WHERE salary > 50000).
- Ask AI to generate equivalent Pandas code.
- Test the generated code on a sample DataFrame.

Expected Output:

- Equivalent Pandas query that retrieves the correct subset of data.

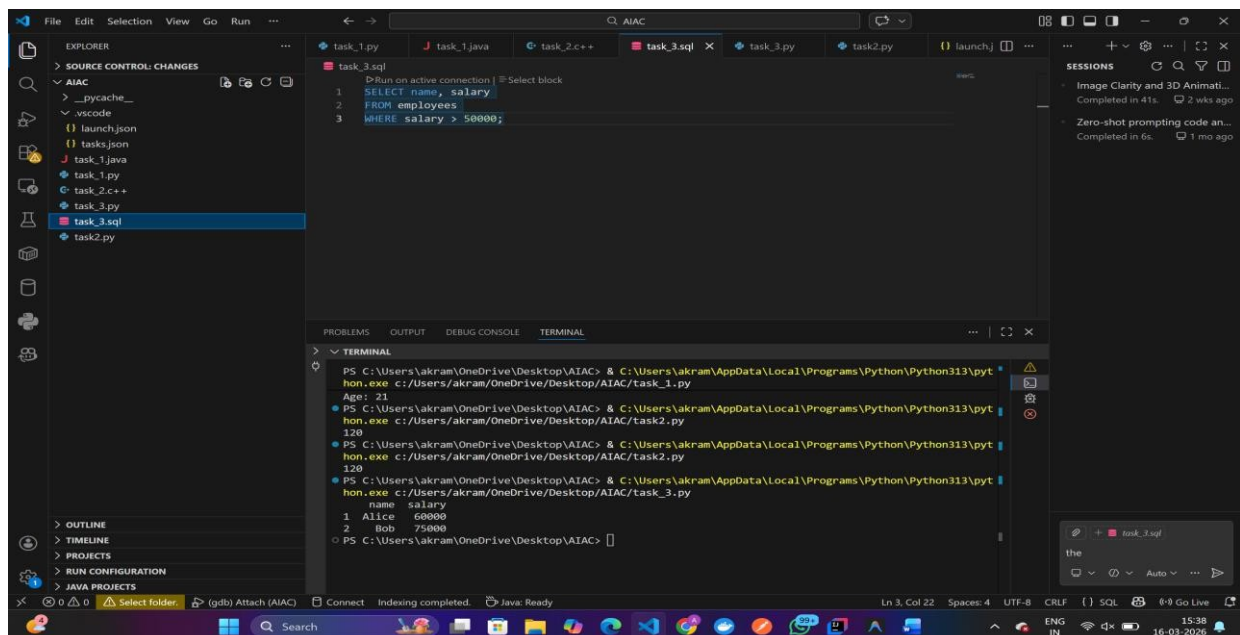
CODE:



```
1 import pandas as pd
2
3 data = {
4     'name': ['John', 'Alice', 'Bob', 'Emma'],
5     'salary': [45000, 60000, 75000, 40000]
6 }
7
8 df = pd.DataFrame(data)
9
10 result = df[df['salary'] > 50000][['name', 'salary']]
11
12 print(result)
```

Terminal Output:

```
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_1.py
Age: 21
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_2.py
120
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_3.py
   name  salary
1  Alice   60000
2   Bob   75000
PS C:\Users\akram\OneDrive\Desktop\AIAC>
```



```
1 SELECT name, salary
2 FROM employees
3 WHERE salary > 50000;
```

Terminal Output:

```
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_1.py
Age: 21
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_2.py
120
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_3.py
   name  salary
1  Alice   60000
2   Bob   75000
PS C:\Users\akram\OneDrive\Desktop\AIAC>
```

Task 4 – Real-Time Application: Multi-Language Snippet

Converter

Scenario:

Students will choose a real-time use case where they need code in multiple languages. Example:

- Converting a Python web scraping snippet into JavaScript for browser automation.
- Translating a Java method into C# for enterprise applications.

Instructions:

- Prompt AI to translate code between two chosen languages.
- Test both implementations to ensure they produce the same results.
- Document challenges faced during translation (e.g., syntax, libraries).

Expected Output:

- Working equivalent code snippets in at least two different programming languages

CODE:

The screenshot shows a Visual Studio Code editor with a Python script named `task_4.py` open. The script uses the `requests` and `BeautifulSoup` libraries to fetch a webpage and extract href values from links. The terminal window at the bottom shows the command to run the script, which has been executed successfully. The output of the script is displayed in the terminal, showing the href values of the links found on the page.

```
1 import requests
2 from bs4 import BeautifulSoup
3
4 url = "https://www.onlinegdb.com/"
5
6 response = requests.get(url)
7 soup = BeautifulSoup(response.text, 'html.parser')
8
9 for link in soup.find_all('a'):
10     print(link.get("href"))
```

```
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_4.py
#editor_1
#
#tab-stdin
#tab-stdout
#tab-stderr
#tab-debug-console
/upgrade-premium
https://www.onlinegdb.com/blog/brief-guide-on-how-to-use-onlinegdb-debugger/
#debug_window_call_stack
#debug_window_local_variables
#debug_window_registers
#debug_window_display_expression
#debug_window_breakpoints
#
#
#
```

